NATIONAL UNIVERSITY OF LESOTHO
Department of Mathematics and Computer Science
CS3431 – Machine Problem 2

Due date: 21st April 2026 before 1500 hours.

Design an AI system that can advise a user on the whether the car he/she wants to buy is in:

1. unacceptable (unacc)
2. acceptable (acc)
3. good (good)
4. very good (v-good)

condition.

The above conditions (classes) are based on the data used to determine them. The data is made up of six (6) attributes namely:

1. buying price (**buying**)
2. price of the maintenance (**maint**)
3. number of doors (**doors**)
4. no. of passengers (**persons**)
5. size of luggage boot (**lug_boot**)
6. car safety (**safety**)

The 6 attributes above are represented by first 6 columns of data contained in the file “car.data”. This data file can be downloaded with from:
https://archive.ics.uci.edu/dataset/19/car+evaluation

Assignment specifications:

(a) Create a **naive Bayes mode**l using *car.data* data .

(b) To create the model, use the Python-based *Scikit-Learn (sklearn)*, packages.

(c) Use tools or other means to split data into **training set** and **test set**.

(d) Train the naive Bayes model with the **training set** and evaluate it with the **test set**.

(e) Once the model is trained and the user is running your program, the user should be able to enter all the car attributes and your system should be able to output the car condition. (e.g. X_test = {v-high, med, 3, 2, small, low} may give output as “acceptable”.)

The assignment should be very easy and straight-forward with Python.

Submit a zipped folder named <YOURSTUDENT NUMBER> .zip or .gz . Make sure that you include a README file that specify how you have trained your model.

The assignment is actually a short quiz. It should be straight forward!!!